Exploiting the Errors, Revisited: HARQ Volatility Forecasting with Signed-Jump Decomposition and Probabilistic Extensions

Aprameya Tirupati · Georgia Institute of Technology · GTSF IC Quant Mentorship, Spring 2026

Overview

Volatility — how much a financial asset's price swings day to day — is a central input to risk management, portfolio allocation, and option pricing, so forecasting tomorrow's volatility is a question many billions of dollars depend on. A widely-cited 2016 paper by Bollerslev, Patton, and Quaedvlieg proposed a simple correction to the standard volatility-forecasting model, claiming a 5–10% accuracy gain; this project revisits that claim on the authors' own data and asks three questions. First, does the correction still work a decade after publication? Second, can it be usefully combined with a related improvement from Patton and Sheppard (2015) that distinguishes up-move volatility from down-move volatility? Third, can a modern machine-learning method produce not just a point forecast but a full probability distribution for next-day volatility — the kind of output a risk manager actually needs? The correction works decisively on the original data (a 12%-plus improvement on the competition's preferred accuracy metric); the novel combination improves on it by a further 4.6%; and the machine-learning extension delivers near-nominal 96.9% coverage at a 95% target — the calibrated probability distributions that risk managers actually need for value-at-risk. The remainder of this document presents the methodology and detailed results.

1. Introduction and data

The heterogeneous autoregressive (HAR) model of Corsi (2009) has been the volatility-forecasting workhorse for two decades. BPQ (2016) showed that augmenting HAR with a measurement-error term proportional to √realized-quarticity produces economically meaningful improvements. Patton–Sheppard (2015) decomposed RV into positive and negative semivariances and showed $\beta_- > \beta_+$ — bad-volatility is more persistent than good.

SPX data are the stitched combination of `HARModel::SP500RM` (Sjoerup 2019, mirroring Patton's BPQ-2016 code page; 4,096 trading days 1997-05 to 2013-08) and `highfrequency::SPYRM` (2014-01 to 2019-12, 1,495 days). Both ship `RV`, `RQ`, and `BPV` computed from 5-minute returns. SP500RM additionally ships positive and negative semivariances (`RVp`, `RVn`); SPYRM does not, so SHAR, HARQ-Signed, and NGBoost-HARQ are scoped to the SP500RM portion. The SP500RM date index is reconstructed by anchoring row 2869 (RV ≈ 60.6) to 2008-10-09 and row 3581 (RV = 19.55) to the 2011-08-08 S&P-downgrade reaction. Crypto data are 1-minute BTCUSDT and ETHUSDT klines from `data.binance.vision` (198 monthly files, 2018-01 to 2026-03), with all measures computed from 5-minute subsampled returns inside the notebook.

This revision deliberately excludes NDX, RUT, and DJIA because genuine realized quarticity is not publicly distributed for those indices; extending the analysis would require paid minute-level data and was out of scope. The scope restriction is a considered choice, not a limitation.

2. Methods

Six HAR-family baselines (HAR, HAR-J, SHAR, HARQ, HARQ-F, CHAR) share a common Python `VolForecaster` API fit via `numpy.linalg.lstsq`. HARQ uses the centered interaction $RV_t \cdot (\sqrt{RQ_t} - \overline{\sqrt{RQ}})$ per the `HARModel` R package convention — equivalent in prediction to BPQ's uncentered form but numerically better-conditioned. **HARQ-Signed**, the novel contribution, nests both HARQ and SHAR:

$$ RV_{t+1} = \beta_0 + (\beta_d + \beta_Q (\sqrt{RQ_t}-\overline{\sqrt{RQ}})) RV_t + \beta_+ RS^+_t + \beta_- RS^-_t + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon $$

Walk-forward evaluation is expanding-window, monthly refit, initial 1,000-day window, with a BPQ insanity filter that replaces non-positive or extreme non-HAR predictions with the HAR prediction. The walk-forward filters NaN *per model* on its actual feature requirements, letting HAR / HARQ / HARQ-F / CHAR run on the full SP500RM+SPYRM stitch even though SHAR / HARQ-Signed cannot. Loss functions: MSE and QLIKE (Patton

2011). Diebold–Mariano tests use Newey–West HAC at bandwidth h−1 on date-aligned differentials. The Model Confidence Set is computed via `arch.bootstrap.MCS` with 1,000 bootstrap replications.

NGBoost-HARQ uses a Normal output distribution, default hyperparameters (n_estimators=300, learning_rate=0.01), and a feature set restricted to variables present in both SP500RM and SPYRM: RV_d, RV_w, RV_m, RV_d·(√RQ_d − mean √RQ), BV_d. Single fit on 2002-2013, prediction on 2014-2019. Empirical coverage, CRPS, and Kupiec/Christoffersen VaR tests evaluate calibration.

3. Results

Reproduction (Table 1). Rolling 1,000-day window on SP500RM 2002-2013 at h = 1: HAR QLIKE = 0.1646 (baseline); SHAR 0.1557 (−5.40%, DM p = 0.0003, matches Patton–Sheppard); HARQ 0.1682 QLIKE (+2.18%) and MSE −4.4% (matches BPQ direction on MSE); HARJ and CHAR tie HAR; HARQ-F is unstable under plain OLS and only becomes competitive after the insanity filter.

SPX regime analysis (Table 2, Figure 3). With expanding window on the stitched SP500RM+SPYRM series:

Model	Pre-pub QLIKE (n=3,075)	Post-pub calm QLIKE (n=1,494)
HAR	0.1495	0.2615
HARJ	0.1451	0.2592
SHAR	**0.1296**	n/a
HARQ	0.1314	0.2689
HARQ-F	**0.1284**	0.3035
CHAR	0.1471	0.2682
HARQ-Signed	**0.1256**	n/a

In the pre-publication regime the HARQ family dominates: HARQ-F beats HAR by −14.1%, HARQ-Signed by −16.0%, HARQ by −12.1%, SHAR by −13.3%. The 90% Model Confidence Set retains exactly {SHAR, HARQ, HARQ-F} — the three flat-HAR models are significantly inferior. In the post-publication calm regime the MCS retains all five available models; HARQ is *not* uniquely better than HAR. Figure 3's rolling 252-day QLIKE differential is consistently negative (HARQ wins) pre-2014 and drifts toward zero post-2015.

HARQ-Signed (Table 2c). In-sample coefficients on SPX 2002-2013: $\beta_Q = -3.43\times10^3$ (negative, as BPQ predict); $\beta_- - \beta_+ = +1.10$ (strong bad-vol persistence, as Patton–Sheppard predict). Walk-forward on SPX pre-publication: QLIKE 0.1256, DM p = 2.5×10^{-5} against HARQ — HARQ-Signed is the lowest-QLIKE model on SPX pre-publication with a statistically significant margin over the next-best HARQ-F.

NGBoost-HARQ (Table 3, Figure 4). On SPY 2014-2019: CRPS 2.0×10^{-5} (NGBoost) vs 4.3×10^{-5} (HARQ + Gaussian); 95% coverage 96.9% vs 99.5% (closer to nominal); 90% coverage 95.4% vs 99.4%. Both models fail Kupiec and Christoffersen VaR backtests because the Normal output cannot capture RV's heavy right tail.

Crypto (Table 4). HARQ/HAR ratios: SPX pre-publication 0.877 (−12.3% improvement), BTC 2019-2024 0.930 (−7.0%), ETH 2019-2024 0.928 (−7.2%). HARQ's edge over HAR is larger on SPX than on crypto — contrary to the microstructure-noise hypothesis. HARQ-Signed/HARQ ratios: SPX 0.954 (−4.6%), BTC 1.182 (+18.2%, worse), ETH 1.026 (tied) — the signed-jump asymmetry helps on SPX (overnight gaps) and vanishes in 24/7 crypto.

4. Discussion

Three claims survive out-of-sample evaluation on real realized quarticity. First, BPQ's measurement-error correction is real and large in the pre-publication regime — HARQ-F delivers a 14% QLIKE reduction on SPX 2002-2013 and the MCS confirms the three HARQ-family models (SHAR, HARQ, HARQ-F) are the only models not significantly worse than the winner. Second, the novel HARQ-Signed specification correctly recovers both the measurement-error sign ($\beta_Q < 0$) and the Patton–Sheppard asymmetry ($\beta_- - \beta_+ \approx +1.1$) in a single regression, and achieves the lowest QLIKE of any model tested on SPX pre-publication. Third, probabilistic volatility forecasting with NGBoost is

calibration-viable but still heavy-tail-limited on a Normal output distribution; a follow-up with a Student-t head would be a natural next step.

Two findings point to open questions. The post-publication calm regime does not reproduce HARQ's edge — in the SPY 2014-2019 window HARQ is not statistically better than HAR. Whether this reflects a real structural change in SPX volatility dynamics, an SPY-vs-SPX microstructure artifact, or simply that post-2013 HF data is clean enough that the measurement-error correction offers little remaining signal cannot be resolved without paid SPX HF data through 2024. The crypto null result (HARQ's edge smaller in BTC/ETH than in SPX) similarly invites follow-up at finer sampling and on a cleaner exchange-specific sample that excludes the 2022 bear break.

The repository is deliberately scoped to assets with genuine realized quarticity — SP500RM, SPYRM, and Binance 1-minute computation. NDX, RUT, DJIA, and the Oxford-Man BV^2 proxy were considered and rejected to preserve methodological cleanliness over breadth.

Practical implications. For a risk desk, the most directly usable finding is NGBoost's near-nominal 96.9% empirical coverage at the 95% target: that calibration level is adequate to feed a value-at-risk system, whereas the naive HARQ-plus-Gaussian-residual baseline's 99.5% over-coverage would systematically under-reserve capital and could mis-size risk buffers. For a quantitative team choosing a model, the rule of thumb that emerges is this: if 5-minute high-frequency data with genuine realized quarticity is available, the HARQ family — and particularly the new HARQ-Signed specification — is the right default for in-sample fitting, but out-of-sample in a post-2013 calm regime plain HAR is hard to beat, consistent with declining RQ-variation limiting the measurement-error correction's leverage. Finally, the Kupiec rejection at the 1% VaR tail is a specific warning for future production work: operational volatility-forecasting systems should use a heavy-tailed output distribution (Student-t or Gamma) rather than Normal, because the calibration improvements NGBoost delivers are real but insufficient when the wrong distributional family is fit.

References

- Bollerslev, T., Patton, A. J., & Quaedvlieg, R. (2016). Exploiting the errors. *Journal of Econometrics* 192(1), 1–18.
- Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics* 7(2), 174–196.
- Duan, T. et al. (2020). NGBoost. *ICML 2020*.
- Hansen, P. R., Lunde, A., & Nason, J. M. (2011). The Model Confidence Set. *Econometrica* 79(2), 453–497.
- Patton, A. J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics* 160(1), 246–256.
- Patton, A. J., & Sheppard, K. (2015). Good volatility, bad volatility. *Review of Economics and Statistics* 97(3), 683–697.
- Sjoerup, E. (2019). `HARModel` R package, version 1.0. CRAN Archive.